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Anomaly Detection Pipeline Architecture

Overview

```
Parse error on line 1:
flowchart TD    INP
^
Expecting 'NEWLINE', 'SPACE', 'GRAPH', got 'ALPHA'
```

Workflows & Subworkflows

Workflow	Subworkflow	Clusterer	Subtypes
numeric	semantic	`SemanticNumericClusterer`	age, rating, year, lat, long, %, temp
numeric	generic	`GenericNumericClusterer`	salary, amount, count
string	validator	`ValidatorStringClusterer`	email, phone, url, uuid, ip, zip, ssn, boolean
string	categorical	`CategoricalStringClusterer`	state, country

Workflow	Subworkflow	Clusterer	Subtypes
string	freetext	`FreeTextStringClusterer`	name, company, address, city, username
string	generic	`GenericStringClusterer`	file_path, product_title
datetime	format	`FormatDateTimeClusterer`	date, datetime
datetime	generic	`GenericDateTimeClusterer`	unknown date formats
identifier	pattern	`PatternIdentifierClusterer`	pattern_id, numeric_id
identifier	generic	`GenericIdentifierClusterer`	unknown ID patterns

Numeric Workflow

```
Parse error on line 1:
flowchart TD    N_I
^
Expecting 'NEWLINE', 'SPACE', 'GRAPH', got 'ALPHA'
```

Semantic Bounds:

- age: 0-120 | rating: 0-10 | year: 1900-2100
- latitude: -90 to 90 | longitude: -180 to 180

Generic Ensemble: IsolationForest + IQR + Z-score (2/3 voting)

String Workflow

```
Parse error on line 1:
flowchart TD    S_I
```

```
^
Expecting 'NEWLINE', 'SPACE', 'GRAPH', got 'ALPHA'
```

Validators: email, phone, url, uuid, ip_address, zipcode, ssn, boolean

Categorical: Frequency-based with typo filtering (80-99% similarity)

FreeText: Lenient - rejects only reserved words + gibberish

DateTime Workflow

```
Parse error on line 1:
flowchart TD    DT_
^
Expecting 'NEWLINE', 'SPACE', 'GRAPH', got 'ALPHA'
```

Format Detection: Identifies majority format (e.g., `%Y-%m-%d`)

Identifier Workflow

```
Parse error on line 1:
flowchart TD    ID_
^
Expecting 'NEWLINE', 'SPACE', 'GRAPH', got 'ALPHA'
```

Pattern Signature: e.g., `AN` for alphanumeric like "C001"

File Structure

```
backend/core/clustering/
├── strategy_selector.py      # Routes subtype → clusterer
├── dynamic_selector.py       # Runtime is_categorical() check
├── base_clusterer.py         # Abstract base class
├── numeric/
│   ├── semantic_clusterer.py # Bounds checking
```

```
│   └── generic_clusterer.py  # ML ensemble
├── string/
│   ├── validator_clusterer.py # Regex validators
│   ├── categorical_clusterer.py # Frequency+fuzzy
│   ├── freetext_clusterer.py  # Lenient
│   └── generic_clusterer.py   # DBSCAN-based
├── datetime/
│   ├── format_clusterer.py    # Format consistency
│   └── generic_clusterer.py
└── identifier/
    ├── pattern_clusterer.py   # Pattern signature
    └── generic_clusterer.py
```

Detection Logic Summary

Clusterer	Detection Method	Threshold
SemanticNumeric	Config bounds (min/max)	Exact
GenericNumeric	IQR + Z-score + IsolationForest	2/3 vote
Validator	Regex pattern matching	100% match
Categorical	Frequency + Fuzzy matching	85% similarity
FreeText	Reserved words + Alpha ratio	30% alpha
DateTime	Format consistency	Majority
Identifier	Pattern signature	Majority